

Nano/microscale roughness control of accident-tolerant Cr- and CrAl-coated surfaces to enhance critical heat flux

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HIGHLIGHTS

- Nanostructure or microstructure alone hardly enhance critical heat flux.
- Hierarchical structures showed a synergetic enhancement on critical heat flux.
- Smaller dry spot controlled by surface capillarity was more advantageous to the enhancement.

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ABSTRACT

This study aimed to improve the thermal safety of accident-tolerant fuel (ATF) cladding by enhancing the pool boiling critical heat flux (CHF) through control of the nano/microscale roughness of ATF-functional Cr- and CrAl-coated surfaces. To diversify the surface structures at the nano/microscale, we ground the surface to achieve the typical roughness range of a nuclear fuel cladding and then deposited the ATF candidate materials of Cr and CrAl on the ground surface. 17 test surfaces were fabricated by grouping three types of surface structures: microstructure, nanostructure, and nano/microstructure. The structural feature was parametrically categorized based on the arithmetic roughness height R_a at microscale and the surface area ratio r_n at nanoscale. While R_a was observed to influence both the nucleate boiling efficiency and CHF, r_n was not capable of enhancing them both exclusively. Nonetheless, a synergistic effect of R_a and r_n on the CHF was observed. The CHF values of microstructure, nanostructure, and nano/microstructure were enhanced by 19%, 9%, and 79%, respectively. A capillary wicking experiment showed that an increase in surface roughness leads to a decrease in the dry area; therefore, a potential increase in the liquid area fraction of the boiling surface contributes to additional evaporation.

1. Introduction

1.1. Background

The primary function of the accident-tolerant fuel (ATF) claddings deployed in nuclear power plant systems is to significantly reduce the high-temperature steam reaction under severe accident conditions, thereby mitigating hydrogen generation [1]. After an elaborate screening of the design strategies applicable to ATF cladding [2,3], coating an oxidation-resistant microlayer on the outer surfaces of existing Zr-based alloys has been identified as a promising method for

short-term development and application. This method has been considered because maintaining the existing Zr-based alloys with a surface coating does not require considerable changes in the neutron absorption efficiency and thermal-hydraulic feedback unlike the changes required when completely replacing the Zr-based alloys.

Incorporating minimal changes in addition to enhancing the oxidation resistance has been recognized as acceptable with respect to the technical requirements of the ATF cladding system. Nevertheless, the research community concerned with enhancing the boiling performance on modified surfaces has reported that a considerable enhancement in the critical heat flux (CHF) in nuclear fuel cladding

Abbreviations: AFM, atomic force microscope; ATF, accident-tolerant fuel; BWR, boiling water reactor; CHF, critical heat flux; CRUD, chalk river unidentified deposits; DNB, departure from nucleate boiling; EDS, energy dispersive spectroscopy; HTC, heat transfer coefficient; PWR, pressurized water reactor; RIA, reactivity-initiated accident; SEM, scanning electron microscope

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Nomenclature

A_b	Bubble base area (m ²)
A_{dry}	dry area (mm ²)
A_{ds}	area of the bubble dry spot (mm ²)
c_{pl}	specific heat of water (J/kg-K)
C_s	surface influencing factor
D_{dry}	diameter of the dry area (mm)
g	gravity acceleration (m/s ²)
h_{fg}	latent heat of water (kJ/kg)
k_l	thermal conductivity of the liquid (kW/m-K)
P_l	liquid pressure (Pa)
P_v	vapor pressure (Pa)
q_w	wall heat flux (kW/m ²)
q_c	critical heat flux (kW/m ²)
q_{bare}	critical heat flux of a bare surface (kW/m ²)
$q_{c,ratio}$	critical heat flux enhancement ratio, q_c/q_{bare} (–)
r_c	cavity radius (m)
r_n	nanoscale surface area ratio (–)
R_{dry}	radius of the dry area (mm)
R_a	arithmetic roughness height (μm)
R_{sm}	arithmetic roughness width (μm)
S_m	mean spacing between the roughness peaks (μm)
T_c	critical temperature (K)
ΔT_{sat}	wall superheat (K)
ΔT_c	critical temperature difference between the critical temperature and the evaporating triple contact angle

Abbreviations

ATF	accident-tolerant fuel
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CHF	critical heat flux
CRUD	chalk river unidentified deposits
HTC	heat transfer coefficient
PWR	pressurized water reactor
BWR	boiling water reactor
RIA	reactivity-initiated accident

Greek symbols

α	dry area fraction (–)
$1-\alpha$	liquid area fraction (–)
γ	influence parameter of the heating surface material
$\theta = \text{MAX}(\theta, 15^\circ)$	maximum contact angle defined by [56]
θ	contact angle (°)
θ_a	apparent contact angle (°)
μ_l	dynamic viscosity of the liquid (Pa-s)
ρ_l	density of the liquid (kg/m ³)
ρ_v	density of the vapor (kg/m ³)
σ_{lv}	surface tension of water (N/m)
δ_l	thickness of the liquid film absorbed inside the surface structure

Subscripts

b	bubble
c	critical
l	liquid
s	surface
sat	saturation
v	vapor
w	wall

applications is achievable [4–7]. As enhancing the CHF on the surface further guarantees an increase in the departure from nucleate boiling (DNB) ratio, it contributes to improving the thermal safety and facilitating the thermal power uprating of the nuclear power plant. A recent RELAP5-3D simulation [8], for instance, showed that even a 10% enhancement in the CHF of ATF claddings could potentially reduce the peak cladding temperature in a reactivity-initiated accident (RIA). This indicates that the ATF cladding surface coating provides an additional opportunity to extend its multifunctional advantages over the scope of the design-basis accidents as well as severe accidents.

Considering all these advantages, several studies [4,9–15] over the last few years have attempted to enhance the CHF of ATF-functional

claddings by altering the structural features at nano/microscale to render the surface characteristics more responsive to wettability and capillary wicking. Using direct current (DC) magnetron sputtering, Seo et al. [12] fabricated FeCrAl-coated layers having different surface roughnesses. They observed that the CHF of the FeCrAl-coated surfaces was enhanced up to 42% and attributed it to an increased roughness factor which augmented the length of the triple contact line for additional rewetting. Son et al. [15] further controlled the intrinsic surface roughness before coating a Cr layer such that the hierarchically augmented roughness induces a capillary action, leading to additional enhancement of the CHF up to 79%. For Cr-coated tube surfaces fabricated by the same technique, Son et al. [14] showed that the CHF

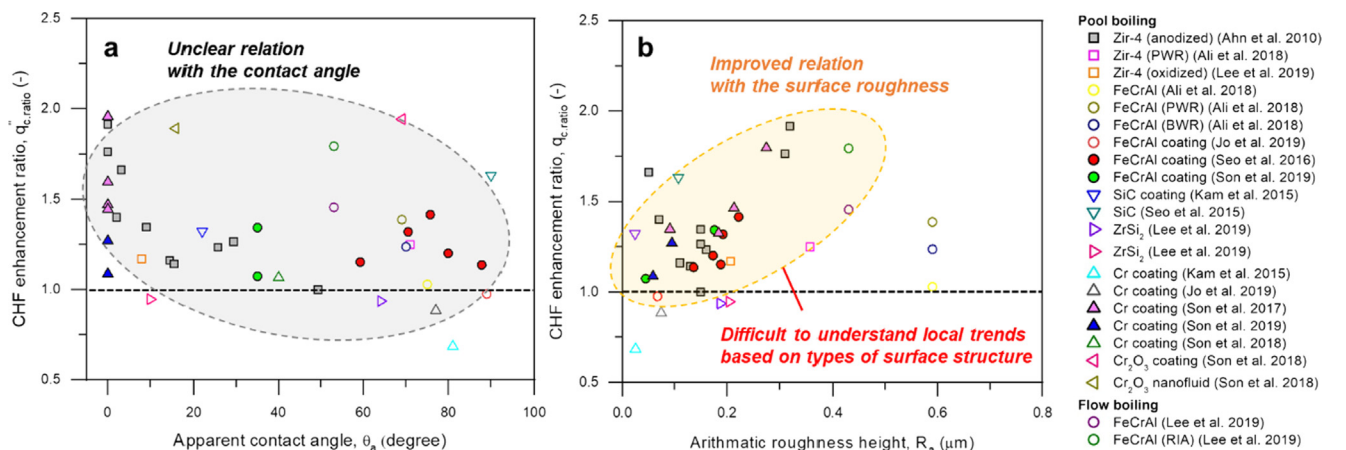


Fig. 1. Global CHF trends of surface-modified Zircaloy-4 claddings [4,9,11] and ATF-functional surfaces [9–12,14,15,25,26] with (a) θ_a and (b) R_a . In these graphs, the open and solid symbols denote the as-received/machined and morphology-controlled test surfaces, respectively.

enhancement ratio can be reduced as the heater orientation is changed from horizontal to vertical in pool boiling. Moreover, Ali et al. [9] observed that improvements in the surface roughness and wettability of an FeCrAl alloy after corrosion under pressurized water reactor (PWR) and boiling water reactor (BWR) chemical conditions enhance the CHF. In contrast, it is difficult to enhance the CHF if the ATF-functional surfaces are finished very smoothly. For example, Jo et al. [16] coated Cr and FeCrAl over a Zr-based surface using a cold spray technique and then polished the coated surfaces again to remove the structural features. For a decrease in the surface roughness up to 10^{-2} μm , they observed no significant change in the CHF.

Besides the surface effect on the CHF, recent quenching experiments especially focused on a Cr-coated surface also show that a change in the surface characteristics affects the quench performance. Lee et al. [17] observed that the quench duration of a Cr-coated niobium (Nb) surface is longer than that of a non-coated surface despite the higher surface roughness and larger heat capacity of the Cr-coated surface. According to Ebrahim et al. [18], the larger the heat capacity, the slower is the transition from film boiling to nucleate boiling. Seshadri and Shirvan [19] evaluated the quench performance of fresh and oxidized Cr-, Fe-CrAl-, and Mo-coated surfaces by measuring the surface wettability and capillary wicking capability. Before oxidation, the quench duration was shortest for the FeCrAl-coated surface, whereas, after oxidation, the surface wettability and capillary wicking of the Cr-coated surface significantly improved, resulting in a better quench performance than the FeCrAl-coated surface. Considering the experimental findings from recent studies, it is obvious that the surface characteristic of the ATF-functional surface is one of key engineering factors capable of affecting both the CHF and the quench.

1.2. Motivation and objective

To define the type of surface parameter that is essential for enhancing the CHF and to describe the issues that still remain unclear, it is necessary to obtain a relation between the surface characteristics and the CHF. Considering that the key surface factors influencing the CHF are known to be surface wettability (quantified by θ_a) and surface roughness (quantified by R_a) [20–22], we collected the available data on CHF, θ_a , and R_a measured on the ATF-functional surfaces and plotted these in Fig. 1. In Fig. 1(a), although the CHF tends to decrease with θ_a , the overall trend with the surface wettability can hardly be understood. Especially, the parametric comparison between the contact angle and CHF is limited to a superhydrophilic region because the contact angle quantity cannot distinguish the surface wettability when $\theta_a = 0^\circ$ [15]. Moreover, the existence of excessive CHF outliers along θ_a aggravates the correlation between the surface wettability and CHF. In contrast, an improved correlation is observed between the surface roughness and CHF: the higher the surface roughness, the higher is the CHF (Fig. 1(b)). In the global context, the positive relation between R_a and the CHF is understood based on the previous findings demonstrating that an increase in the surface roughness augments the liquid/solid contact lines [23] and provides more capillary channels [24], thereby supplying more cooling liquid to evaporate the contact lines.

However, an evaluation of the CHF for various surface roughnesses defined at different scales (i.e., nano-, micro-, and nano/microscales) is still lacking toward understanding the local behavior of the surface roughness with respect to the CHF. This is because most of the previous studies tend to introduce as-received/as-machined ATF surfaces [9–11,13,25,26] for the evaluation of the CHF rather than systematically defining the fabrication parameters during the polishing, grinding, and deposition processes for nano/microscale roughness

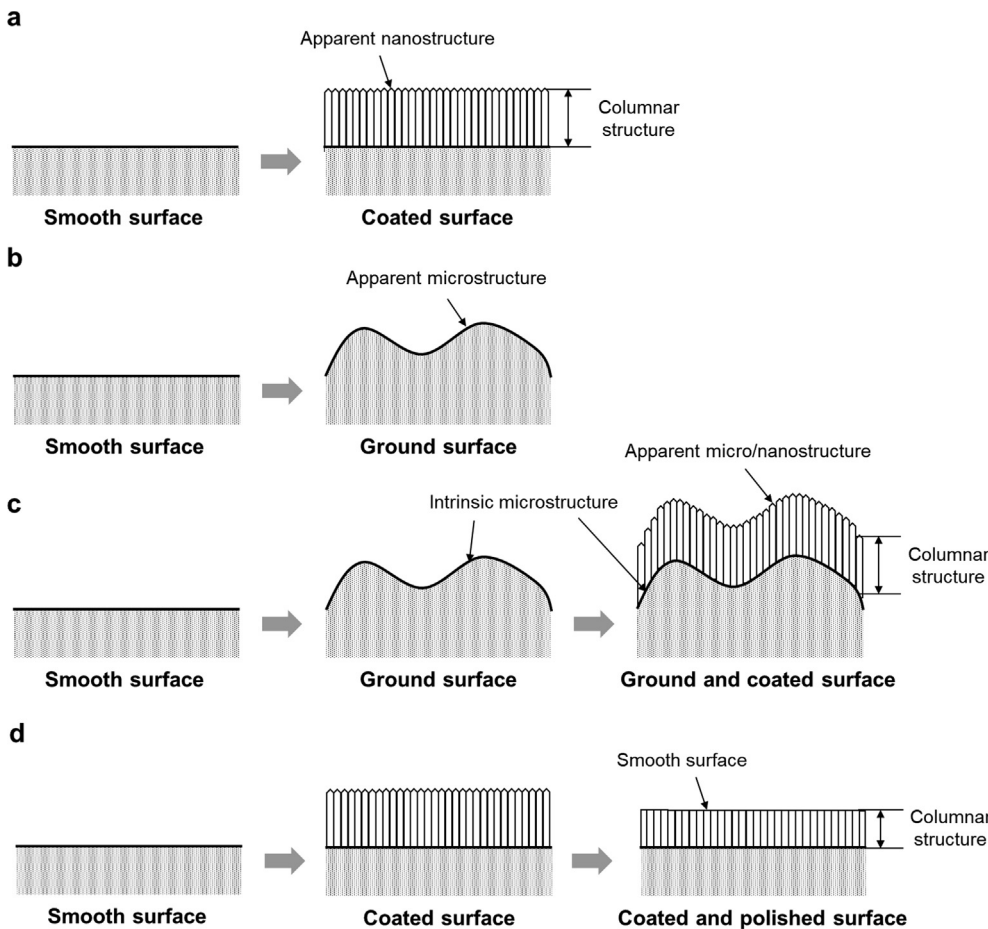


Fig. 2. Schematic classification of the structural features with stepwise fabrication of the surface: (a) nanostructure, (b) microstructure, (c) nano/microstructure, and (d) smooth structure. Note that since the columnar structure inside the coated layer is not porous, the surface characteristics are determined only by the apparent structure.

control. Although a few studies have reported the CHF of roughness-controlled surfaces for ATF application by polishing Cr- and FeCrAl-coated surfaces [16], controlling the deposition process of FeCrAl [12], and controlling the grinding/deposition processes of Cr [15] and Cr and FeCrAl [14], these studies provide a limited understanding of the separate effects of the surface roughness with respect to the nano- and micro-scales. Thus, it is necessary to establish a design basis for the ATF cladding surface in need of nano/microscale roughness control. Moreover, it would potentially assist ATF cladding developers in understanding how controlling the surface roughness affects the CHF.

In this regard, this study experimentally investigates the effect of multiscale roughness of the ATF-functional surface on the CHF and clarifies which roughness scale and combination are effective in enhancing the CHF. We control the multiscale roughness by either polishing or grinding a substrate surface and subsequently depositing the ATF candidate materials (Cr and CrAl are selected for this study) over the surface. After grouping the fabricated surfaces according to the surface roughness defined at nano-, micro-, and nano/microscales, the effect of the different surface roughness scales on the CHF is investigated. Moreover, we characterize the capillary wicking capability such that it is possible to observe the contact line behavior under room condition, which would thereby indicate the underlying cause of the CHF being different for each surface.

2. Multi-scale roughness control at the level of the nuclear fuel cladding surface

2.1. Stepwise fabrication of the surface structures at nano/microscale: Surface grinding and coating

Various types of surface structures (e.g., nano-/micro-roughened [12–15,24], nano-/micro-porous [27–29], and nano-/micro-patterned [21,30–32]) have been introduced in the past to improve the pool boiling performance. However, it is challenging to apply porous and patterned surfaces to nuclear fuel cladding despite their considerable benefits in boiling enhancement. This is because these structures are vulnerable to surface aging due to the significantly enlarged surface area ratio at nano/microscale. In addition, it is technically challenging to fabricate their high-scale structures with the ATF materials. Therefore, the remaining options have been considered in this study. Based on the illustrations shown in Fig. 2, the structural features of each surface and their potential effects on the CHF are briefly explained as follows.

- **Type 1 (Fig. 2(a)):** The ATF materials are uniformly deposited on a smooth surface, thereby finishing the apparent surface at nanoscale without any microscale features. This type of surface structure is unlikely to affect the CHF because the nanoscale roughness seems to be too low to remarkably improve the capillary action.
- **Type 2 (Fig. 2(b)):** The conventional method of grinding the surface generates a considerable number of unidirectional grooved lines, each of which plays the role of a capillary channel [15,24]. This surface finishing is similar to those being performed in existing nuclear fuel claddings. The apparent structure demonstrates a low microscale feature with a roughness height (R_a) < 1.0 μm .
- **Type 3 (Fig. 2(c)):** By depositing the ATF materials over the intrinsic microstructure in Type 2, the apparent feature of the surface becomes hierarchical at nano/microscale, which results in an increase in the surface area ratio to a value higher than those of the other types. As an increase in the surface area ratio amplifies the liquid/solid adhesive or capillary forces [32,33], it is expected that the CHF is more enhanced than those in the other types.
- **Type 4 (Fig. 2(d)):** The surface of Type 1 is smoothened by polishing the apparent nanostructure. This type of surface has been recognized as one of the most promising for ATF cladding applications because the smooth surface minimizes the area that is in contact

with the primary coolant, thereby mitigating surface oxidation [34]. In contrast to the possible gain in terms of the integrity of the surface, the structural effect of the smooth feature on the CHF is negligible regardless of the surface wettability [21,35]. Likewise, Jo et al. [16] observed that polishing Cr- and FeCrAl-coated surfaces does not lead to any changes in the CHF despite a slight variation in the surface wettability.

Among the abovementioned surface types, we adopted Types 1, 2, and 3 and did not consider Type 4 as its smooth surface would not be effective for CHF enhancement. A modification process in two steps was performed to categorize the scales of the apparent structures into nano-, micro-, and nano/microscale. First, to generate different intrinsic microstructures before coating the surface with ATF materials, the test surfaces were ground with different grits of sandpaper. The purpose of this process was to limit the roughness range to an engineering scale applicable to nuclear fuel cladding, in which the surface oxidation under normal operating conditions is not an issue because the surface roughness is close to the technical standard ($R_a \approx 0.3 \mu\text{m}$) of the Zr-based alloys in commercial use. Typically, a higher grit number (having a lower particle diameter) leads to a lower surface roughness. Therefore, relatively high grits of sandpaper, including 800, 600, 400, and 320, were considered. Moreover, the surface was carefully ground along a single direction to generate unidirectional scratches, which provided abundant grooved channels at microscale. Based on Darcy's law that accounts for the permeable characteristic of a liquid along a porous medium [36], these grooved channels can drive the capillary flow because microscale flow channels bear small cross-sectional areas, which can cause a viscous pressure drop for a specific distance [14,15,24].

The second step was to generate nanostructures over the intrinsic microstructure. Among the ATF candidate materials such as FeCrAl, Cr, CrAl, and SiC, we considered Cr and CrAl as the coating materials. Experimentally, Cr- [37,38] and CrAl- [39] coated Zr-based alloys have guaranteed an excellent adhesion integrity even under high-temperature steam oxidation. Owing to the poor adhesion strength of FeCrAl and SiC coatings with Zr-based claddings, these materials have been considered as alternatives to replace the cladding body itself rather than being used as a protective coating [3]. Two coating techniques, DC magnetron sputtering and arc ion plating, were performed to deposit Cr and CrAl on the ground surfaces, respectively. These coating techniques demonstrate a high degree of freedom for the use of the metallic material as the coating target. All the coating conditions of Cr and CrAl are summarized in Table 1.

During the DC magnetron sputtering of Cr, the growth mechanism of the metal film depends on the sputtering parameters such as substrate temperature, inert gas pressure, exposure time, and DC power [40]. Among these parameters, the substrate temperature significantly affects the diffusion of adatoms, and this process plays a role in determining the size of the agglomerated clusters. Typically, a high substrate temperature actuates the diffusion of adatoms, which, in turn, forms compact Cr clusters at nanoscale. This implies that controlling the substrate temperature contributes to noticeable changes in the apparent nano/microstructures. Within the allowable operation limit of the sputtering facility, the substrate temperature was set to 25 $^{\circ}\text{C}$, 150 $^{\circ}\text{C}$, 300 $^{\circ}\text{C}$, and 600 $^{\circ}\text{C}$ to diversify the morphology and roughness of the coated surface.

The CrAl-coated surface was fabricated by the ATF Technology Development Division at the Korea Atomic Energy Research Institute. The coating developers at the institute determined the relative composition of the CrAl alloy by testing its effect on the ATF cladding primary functions (i.e., high-temperature oxidation resistance and mechanical strength). Representatively, they reported that coating the CrAl alloy (Cr 85 wt%: Al 15 wt%) on a Zr-based cladding enables the coated surface to be more resistant to high-temperature steam conditions as well as normal operating conditions [41]. Under oxidation

Table 1
Summary of the surface coating conditions.

Coating parameters	Coating technique	
	DC magnetron sputtering	Arc ion plating
Target material	99.95% pure Cr	CrAl (Cr 80 wt%: Al 20 wt%)
Substrate material	Stainless steel grade 304	Stainless steel grade 304
Substrate temperature (°C)	25, 150, 300, 600	250
Exposure time (h)	1	20
Base pressure* (Torr)	1×10^{-5} (1.33×10^{-3} Pa)	1×10^{-5} (1.33×10^{-3} Pa)
Working pressure (Torr)	1×10^{-2} (1.33 Pa)	2×10^{-2} (2.66 Pa)
DC power (W)	150	4,800
Sputtering gas	Ar	Ar
Ar flow rate (sccm)	29.7	n/a
Pre-heating	30 min. at the corresponding substrate temperature	–
Post-heating	30 min. at the corresponding substrate temperature	–

* This indicates the vacuum pressure held before the Ar injection, thereby significantly reducing contaminations inside the chamber.

conditions, CrAl forms Al_2O_3 over the coated surface, and so the protective layer consisting of Al_2O_3 significantly delays further oxidation [42]. In this study, we used a single composition of the CrAl alloy (Cr 80 wt%: Al 20 wt%), which has been verified by the KAERI, thereby focusing only on its surface characteristic. By setting the power condition in the arc ion plating process 32 times higher (4,800 W) than that of DC magnetron sputtering (150 W), a surface roughness approximately seven times higher was obtained in the CrAl-coated surface. A high-scale roughness of this surface could not be achieved with DC

magnetron sputtering due to the capacity limitation of the power supply. Further details on the arc ion plating technique are provided in [38]. While DC sputtering technique exhibits an inherent disadvantage of coating a large area, other relevant fabrication techniques such as PVD and cold spray coating may be applicable for the practical large dimension of the nuclear fuel cladding. Feasibility study of applying these techniques will be valuable in diversifying their applicability in near future.

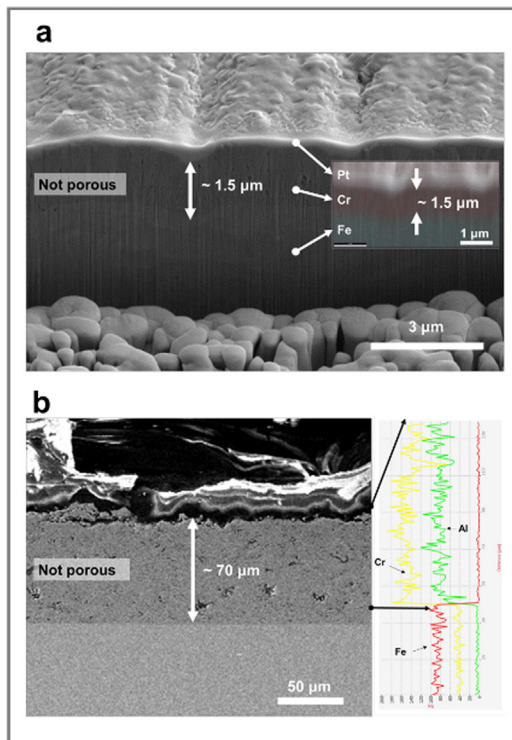
2.2. Observation of the coated layer and surface morphology

After 1 h of sputtering, a Cr layer of approximately 1.5 μm thickness was formed with fine columnar grains (Fig. 3(a)). The mapping result of energy dispersive spectroscopy (EDS) also indicates that an interlayer between the Pt and Fe layers comprises a relatively higher composition of Cr compared to others. Fig. 3(b) shows that a thick CrAl layer of approximately 70 μm thickness was formed during the arc ion plating process. Using the EDS line detection method, it was further confirmed that Cr and Al were mainly distributed inside the CrAl layer, while Fe rarely existed. It is important to note that both the Cr and CrAl layers were not porous except for isolated microdefects along the CrAl layer. Therefore, it is evident that the boiling performance depends only on the 'surface' characteristics (e.g., roughness and wettability) for a particular thickness of the coated layer.

As shown in Fig. 3(c) and (d), the compositions of the major coating elements measured on the Cr- and CrAl-coated surfaces were almost similar to the design values shown in Table 1. Although the oxygen of 2.29 wt% was measured on the Cr-coated surface, its contribution to the total amount seems negligible compared to the dominant contribution of the Cr of 97.71 wt%. Likewise, on the CrAl-coated surface, the actual composition of the CrAl coating (82.92 wt%: Al 17.08 wt%) changed slightly from the design value of the CrAl alloy (Cr 80 wt%: Al 20 wt%).

The SEM images obtained with various magnifications are shown in

Coated layer



Coated surface

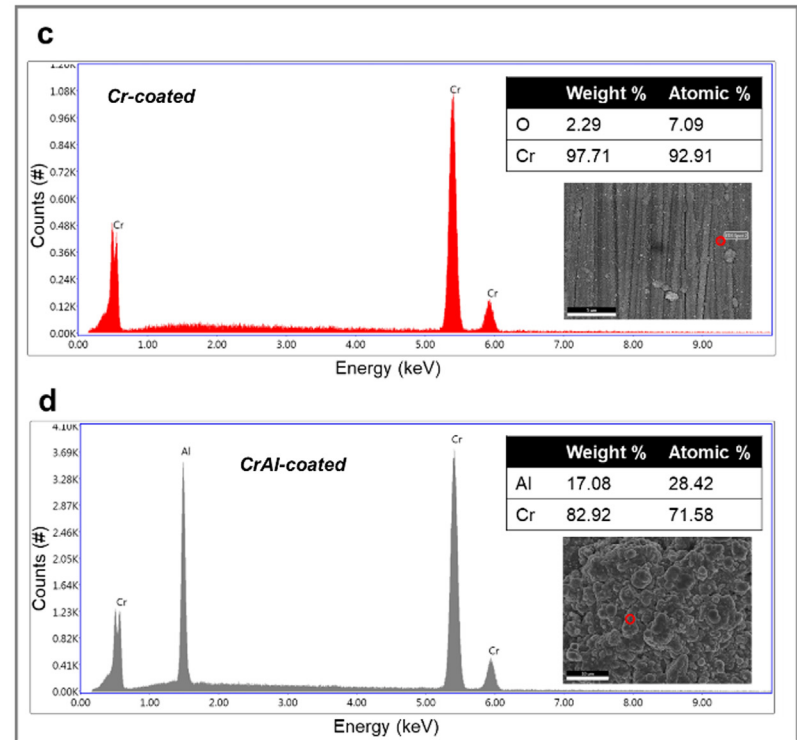


Fig. 3. Observation of (a) Cr- and (b) CrAl-coated layers with the scanning electron microscope (SEM) images and its quantitative identification using the EDS line detection method. To visualize the cross sections of each sample, the Cr-coated thin film was milled by a focused ion beam, while the CrAl-coated thick layer was cut with a low-speed diamond cutter. (c) and (d) show the compositions of the major elements on the Cr- and CrAl-coated surfaces, respectively.

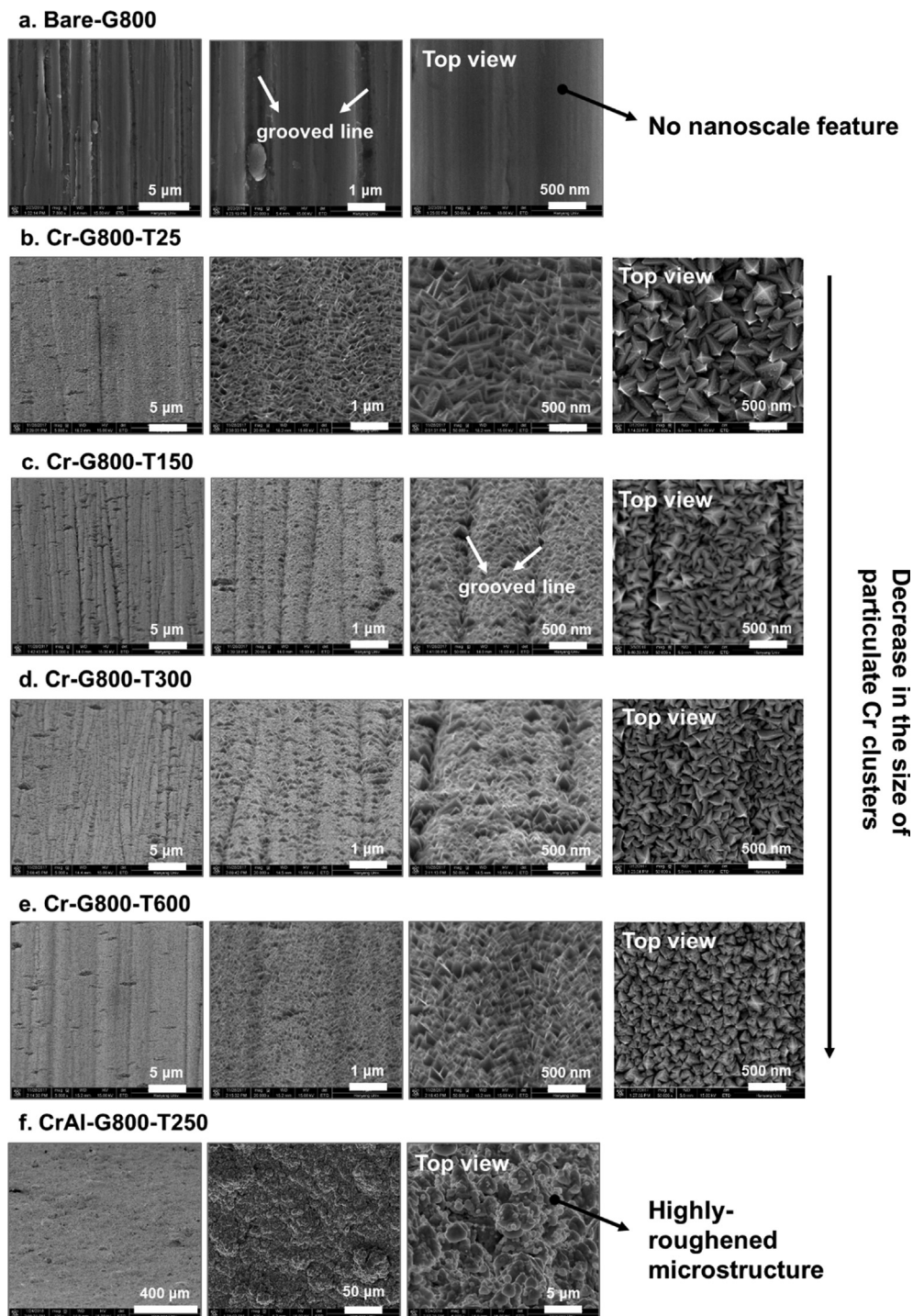


Fig. 4. SEM images of (a) Bare-G800 and 10°-tilted (b) Cr-G800-T25, (c) Cr-G800-T150, (d) Cr-G800-T300, (e) Cr-G800-T600, and (f) CrAl-G800-T250. (a)–(e) show the SEM images with various magnifications, including $\times 5000$, $\times 20,000$, and $\times 50,000$. The magnification of (f) is $\times 100$, $\times 500$, and $\times 5000$ in that particular order. In designating the sample ID, G and T denote the grit number of the sandpaper and the substrate temperature for surface coating, respectively.

Fig. 4. Here, the morphological differences between bare, Cr-, and CrAl-coated surfaces are observed. While the bare surface only exhibits the microgrooved lines without nanostructures, the Cr-coated surfaces demonstrate particulate nanostructures of Cr cluster along the grooved lines. In particular, the size of the Cr cluster tends to decrease as the substrate temperature increases. This implies that the nanoscale surface area ratio can be varied by controlling the substrate temperature. From Fig. 4(f), it is observed that the high-power arc ion plating (4,800 W) process generates a rougher structure at microscale than the low-power DC magnetron sputtering (150 W) process. The qualitative observation

of the surface morphology provides a rationale that the structural features of the Cr- and CrAl-coated surfaces vary between the nano- and microscales. Therefore, these roughness values at each scale are quantified in Section 3.1.

3. Experimental procedure

3.1. Measurement of surface roughness and wettability

For a quantitative analysis of the morphological changes, the

surface roughness was measured by an atomic force microscope (AFM) (XE-100 model) in which the noncontact mode was used for facilitating a high-resolution nanoscale measurement (≈ 0.01 nm). To distinguish the raw configurations of the surface roughness values between the nano/microscales, the roughness profiles were obtained at different AFM resolutions: $30 \times 30 \mu\text{m}^2$ and $5 \times 5 \mu\text{m}^2$ for micro- and nanoscales, respectively (Fig. 5). Similar to the observations in the SEM images, the profile at microscale (Fig. 5(a)) is observed to be the roughest on the CrAl-coated surfaces among all the coated surfaces. The surface roughness values of the bare and Cr-coated surfaces are approximately two times lower than that of the CrAl-coated surface. The difference in the microscale roughness profiles of the bare and Cr-coated surfaces is almost negligible. At the nanolevel (Fig. 5(b)), a frequent peak-to-peak variation along the roughness profile is observed on the Cr-coated surface; however, such a variation is hardly observed on the other surfaces. Among the various roughness parameters, we measured the arithmetic roughness height R_a and the nanoscale surface area ratio r_n at the AFM resolutions of $30 \times 30 \mu\text{m}^2$ and $5 \times 5 \mu\text{m}^2$, respectively. Here, R_a and r_n physically mean the average of the absolute height values along the sampling length and the ratio of the actual area over the projected area, respectively. The mathematical expression of each parameter is shown in Fig. 5.

Importantly, as the roughness features cannot be perfectly uniform over the ground and sputtered surfaces, using the roughness value measured only at a single position may be too irrational to signify the representative nature of the surfaces. To avoid this overgeneralization, the uniformity of the micro/nanoscale surface roughness was evaluated with repeatable measurements at different positions. As shown in the inset histogram in Fig. 6(a), the measurements were repeated between as few as three times and as many as nine times. The repeatability error was calculated by dividing the standard deviation by an averaged value. Fig. 6(b) shows that the repeatability error in R_a is under 22%, which is higher than the 6% maximum error in r_n . These repeatability errors are considered during the comparison of the CHF data in Section 4.2. Considering that the uniformity of the grooved lines at microscale is relatively lower than that of the nanoscale surface area ratio, introducing a reliable engineering process to regularly pattern the microgrooved lines over the metal surface is necessary in the future.

The overall trends of the measured R_a and r_n are shown in Fig. 6(b).

First, R_a increases as the grit number of the sandpaper decreases. When the same grit number is used, R_a hardly varies in spite of changing the substrate temperature for the sputtering process. As an exceptional case, the surface roughness of CrAl-G800-T250 increases significantly up to 7.8 times compared to the Cr-G800-T25/150/300/600 cases. This result can be attributed to the fact that a sputtering power 32 times higher than that of the other cases was applied for the CrAl coating. Second, a decrease in the substrate temperature causes an increase in r_n (except for the cases of Cr-M-T600 and Cr-G800-T600) and scarcely affects R_a . This signifies that controlling the sputtering conditions dominated the changing nanoscale features more than augmenting the microscale roughness. Based on the observed trends, it is evident that the grinding and coating processes enable the categorization of the surface structure into micro- and nanoscales, respectively.

In addition to the surface roughness, a contact angle was measured to quantify the surface wettability. Fig. 7(a) shows the measurement system of the contact angle, which includes an EasyDrop 100 goniometer. A water droplet having a volume of 5 μL was first generated at the end of the syringe. To exclude the effect of impingement of the droplet with the surface while measuring the contact angle (which causes unintended rebound of the droplet [44]), the test surface slowly moved upward and touched the water droplet. Instantly, the liquid/solid contact line began spreading over the surface and stopped at its equilibrium state. After confirming that the contact line did not move anymore, the apparent contact angle θ_a was measured by the imaging system of the EasyDrop 100 goniometer.

Fig. 7(b) shows that, after coating Cr and CrAl, the surface wettability changes from hydrophilic ($60\text{--}70^\circ$) to superhydrophilic ($\approx 0^\circ$). To check which of the roughness parameters out of R_a and r_n mainly affect θ_a , we compare the relations in Fig. 7(c) and (d), respectively, with the available data of R_a , r_n , and θ_a obtained from the studies that report a change in the wettability due to modifications in metallic surfaces. While the effect of R_a on θ_a seems to be unclear (Fig. 7(c)), either oxidation, anodization or an increase in r_n enables the surface to be more hydrophilic (Fig. 7(c) and (d)). Oxidation [45] increases the liquid/solid surface adhesion and the existence of the nanostructure [15,33] enlarges the nanoscale surface area; both of them play a role in further attracting the liquid to spread over the surface. Similarly, it is observed that on differently roughened surfaces, θ_a hardly varies with

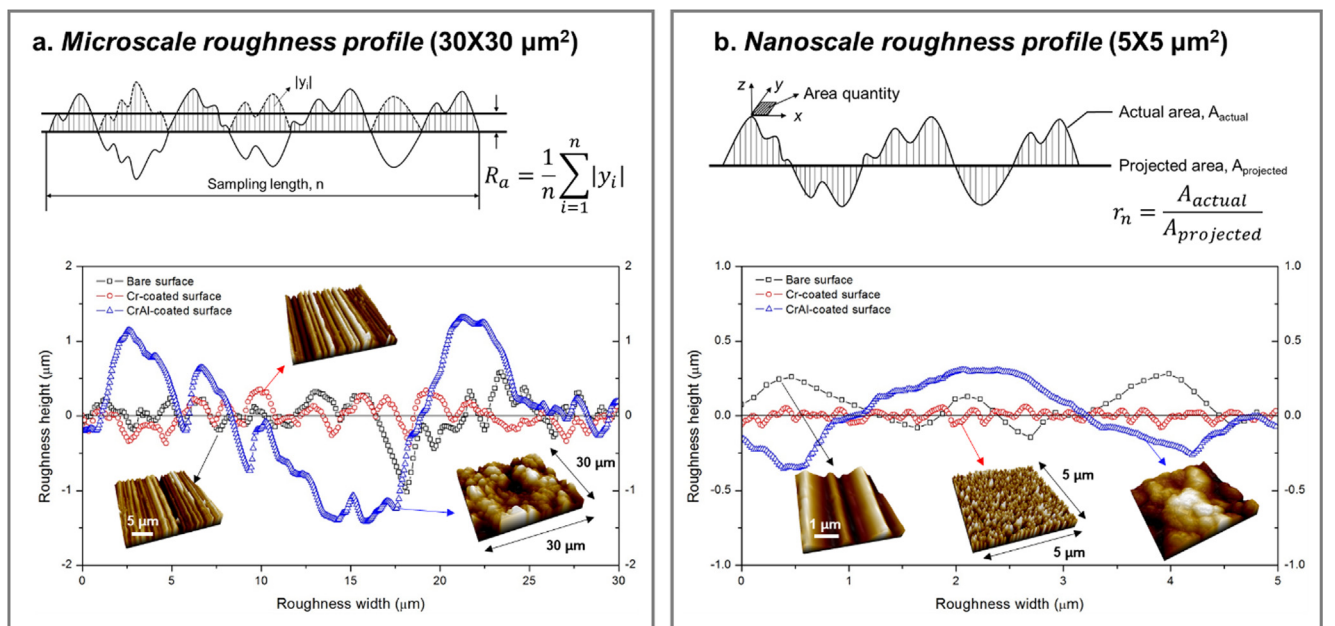


Fig. 5. Comparison of the representative roughness profiles measured from bare, Cr-coated, and CrAl-coated surfaces. The measurement scales of the roughness profiles are categorized into (a) microscale ($30 \times 30 \mu\text{m}^2$ AFM resolution) and (b) nanoscale ($5 \times 5 \mu\text{m}^2$ AFM resolution).

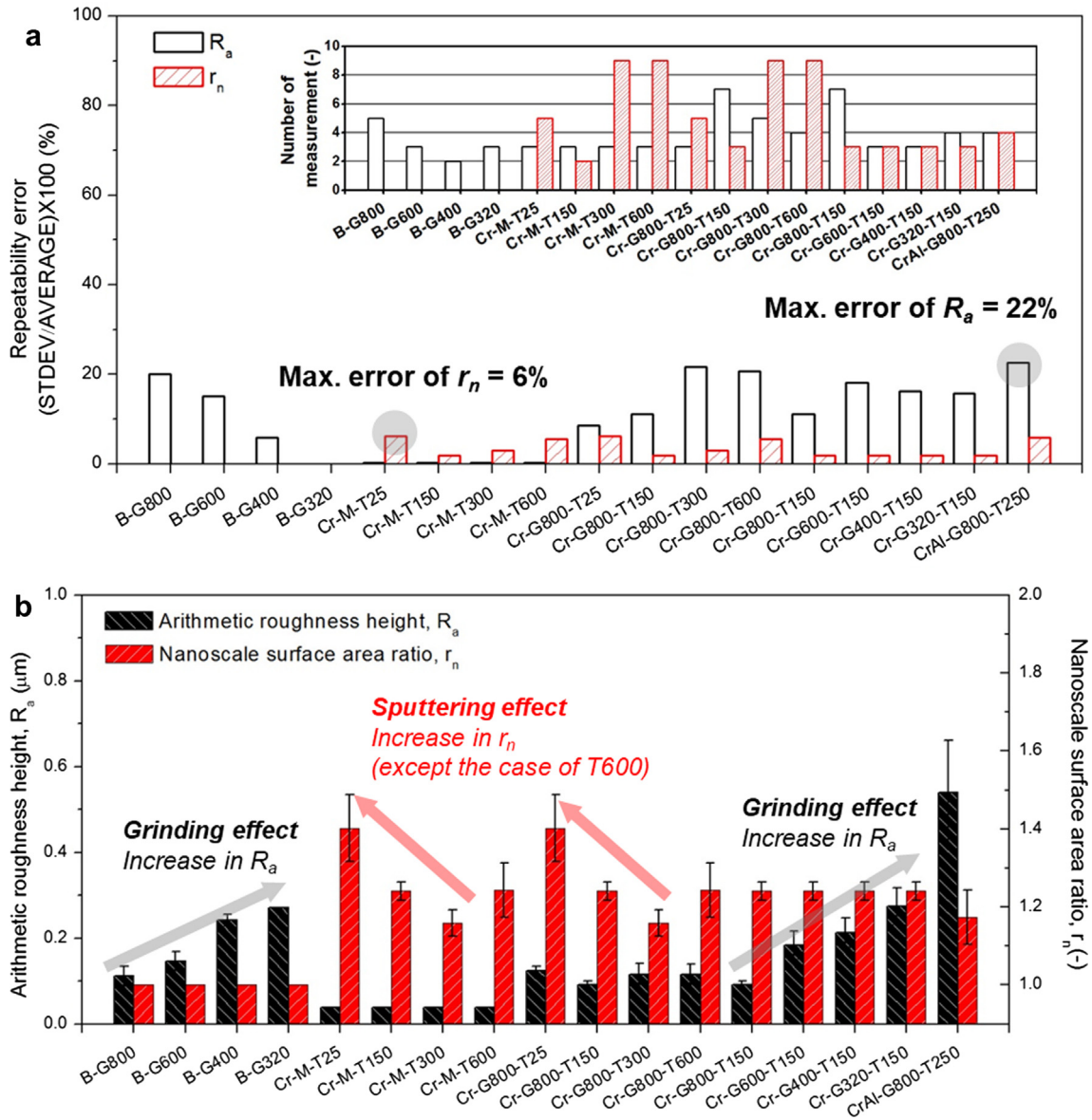


Fig. 6. (a) Iterative measurement of R_a and r_n and their repeatability errors. (b) Effects of the grinding and sputtering processes on the variation of R_a and r_n . In (b), r_n of Cr-M-T600 and Cr-G800-T600 hardly agree with the trend of the sputtering effect. These exceptions could be due to the recrystallization of adatoms, which is accompanied by high surface mobility and bulk diffusion at high substrate temperatures [12,43].

changes in R_a [24], whereas the surface wettability is significantly improved by an augmentation of the nanoscale feature rather than a variation in R_a [28]. Therefore, it is deduced that the superhydrophilicity of the coated surfaces is mainly the result of the generation of the nanostructures.

3.2. Measurement of the capillary wicking capability

There are several ways to characterize the surface capillarity, i.e., capillary rise [46,47], hemi-spreading [48], wickability [32,49], and receding capillary flow [50]. Among these, we used the characterization method of the receding capillary flow, which is capable of mimicking the *in-situ* capillary wicking behavior [50,51]. As described in our previous study [50], this method considers the contribution of the vapor recoil pressure to the receding motion of an *in-situ* triple contact line. It enables the measurement of the size of an expanding dry area that varies with the nano/microscale surface roughness and wettability. With respect to the CHF, a smaller size of the dry area indicates a higher liquid area fraction occupied along the bubble footprints [52], which

implies that more heat flux is necessary to bear the additional evaporation.

Fig. 8(a) shows the experimental apparatus; it consists of an air supply line (connected to an air compressor), a syringe needle for air impingement (inner diameter of 0.52 mm), a test assembly, a high-speed camera, and a light source (both of which are located outside the photograph). The design values for the experiment are specified in Fig. 8(b). In the experimental procedure, we first spread sufficient deionized water ($\approx 20^\circ\text{C}$) to entirely cover the test surface. Over the wet surface, a jet of air compressed to 0.4 MPa was shot from the syringe needle. Then, the liquid competed with the air jet drying the surface such that the pressure of the jet rapidly drove the flow of the bulk liquid that is not captured by the surface structure, while the liquid that is absorbed inside the surface structure slowly receded against the driving pressure.

A visual observation shows an extremely different drying behavior between the bare and coated surfaces (Fig. 8(c)): the bare surface is rapidly dried out without any wicking region, whereas the coated surface that is capable of wicking the liquid resists being totally dried

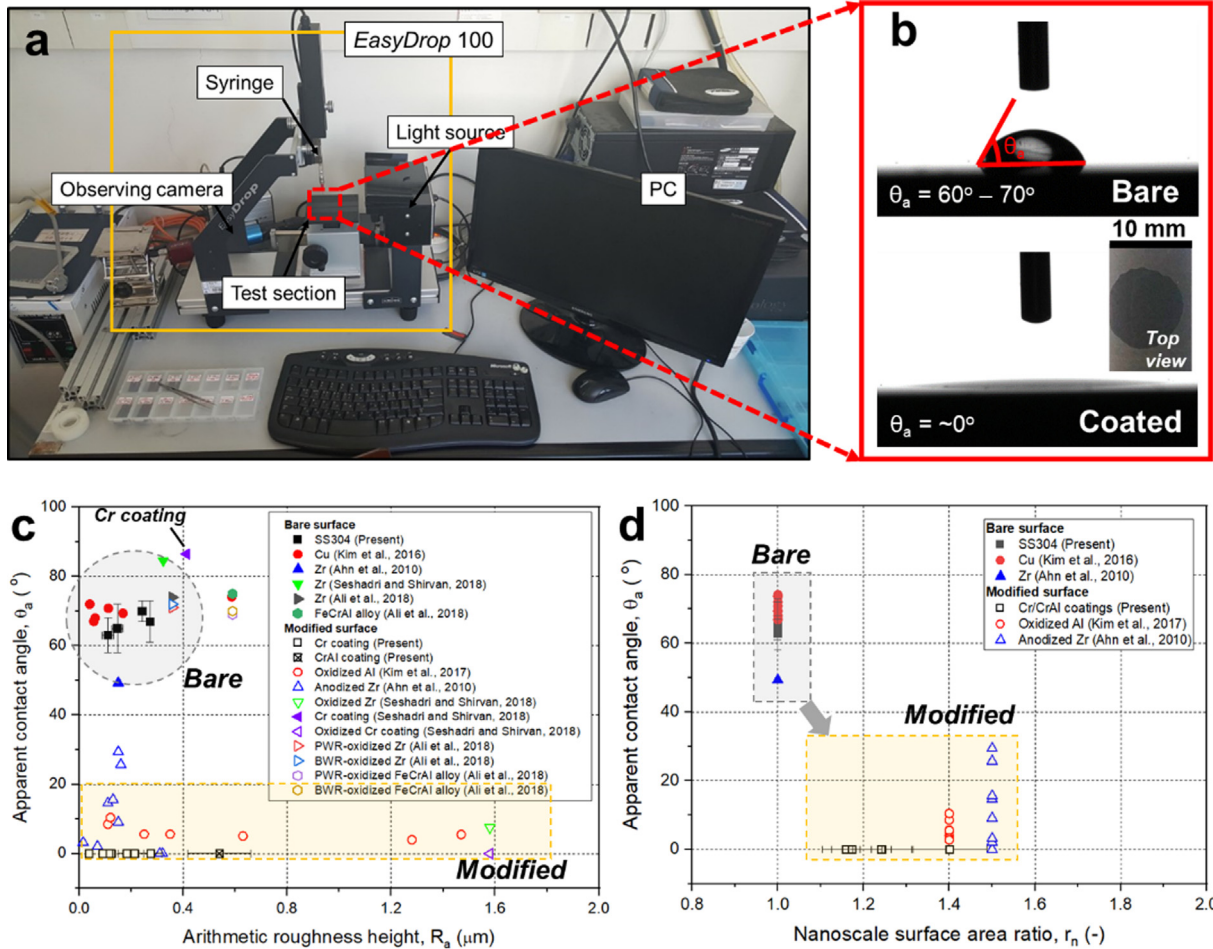


Fig. 7. Measurement of the apparent contact angle. (a) A photograph of the contact angle measurement system. (b) Representative images of the apparent contact angle on the bare and coated surfaces. (c) Ineffectiveness of R_a and (d) effectiveness of r_n on θ_a . In (c), r_n of [28] and [4] were estimated to be 1.4 and 1.5, respectively, based on the SEM images. Note that the oxidation of Cr and CrAl layers under normal operating and accident conditions forms oxide layers of Cr_2O_3 for the Cr layer [19,38] and $\text{Cr}_2\text{O}_3/\text{Al}_2\text{O}_3$ for the CrAl layer [41,42], which could contribute to the surfaces being more wettable [19].

out by further competing along the liquid/solid/air contact line. Here, since the major focus is on comparing the delayed growth behavior of the dry area between the roughness-controlled surfaces, we quantified the expansion rate of the dry area, dA_{dry}/dt , and calculated the dry area, A_{dry} , using a reference time of 5.3 ms at which the diameter of the dry area of a bare surface reaches a capillary length $L_c = \sqrt{\sigma_{lv}/\rho_l g} = 2.5\text{ mm}$ for saturated water. The quantified data of dA_{dry}/dt and A_{dry} are shown in Table 2, and the result is discussed in Section 4.3.

Based on the measured data with respect to the roughness parameters, apparent contact angle, and capillary wicking capability, a test matrix is organized in Table 2. There were 17 cases in all, which were categorized under three types of structural features, i.e., microstructure, nanostructure, and nano/microstructure. The microstructure group, in which the surfaces were not coated, was used to investigate the effect of the microscale roughness R_a on the CHF. The role of this group was to provide the bare CHF data of the intrinsic microstructure for comparing the CHF gain on the coated surfaces. The nanostructure group, in which the surfaces were not ground, was used to investigate whether the nanoscale surface area ratio r_n , excluding the microscale features, could enhance the CHF. The results can reveal the effects of controlling the sputtering conditions on the CHF gain. Finally, the nano/microstructure group, in which the surfaces were ground and coated, was used to investigate the synergetic effect of the microscale roughness and nanoscale surface area ratio on the CHF. The major focus of this group was to estimate the amount of CHF enhancement that could be achieved compared to the other groups.

3.3. Measurement of the pool boiling curve up to CHF

The design of the pool boiling apparatus shown in Fig. 9(a) is similar to that used in our previous study [15]. Its design for this study was slightly modified to incorporate some changes in the insulation of the test section (Fig. 9(b)), which generated a different boiling trend between bare surfaces. The pool boiling chamber (shown in Fig. 9(a)) made of a polycarbonate was filled with deionized water of high electrical resistivity (18.5 M Ω -cm), which ensured that the electricity leakage through the deionized water was negligible. Four immersion heaters were consistently operated to retain the deionized water saturated under atmospheric pressure. Before supplying the DC power into the test section, the saturated water was degassed, until non-condensable gases were invisible inside the pool boiling chamber. Fig. 9(b) shows the assembling of the test section with a test sample: a flat plate of 45 mm length, 10 mm width, and 2 mm thickness. Out of the total length of 45 mm, the heating length was only 25 mm as 10 mm was used for electrical contact with the copper holders at both ends. Thus, the effective heat transfer area was $25 \times 10\text{ mm}^2$. Importantly, the 5 mm gap between the test sample and the supporting mount made of polyetheretherketone (PEEK) was filled with a liquid epoxy (a mixture of SaplKwick™ 20-3566 and 20-3568), which was completely solidified after approximately 2 h. The insulation obtained through this solidification of the liquid epoxy guarantees a larger contact area with the test sample than a solid-to-solid contact between PEEK and the test sample, thereby reducing the heat loss through the narrow gaps around

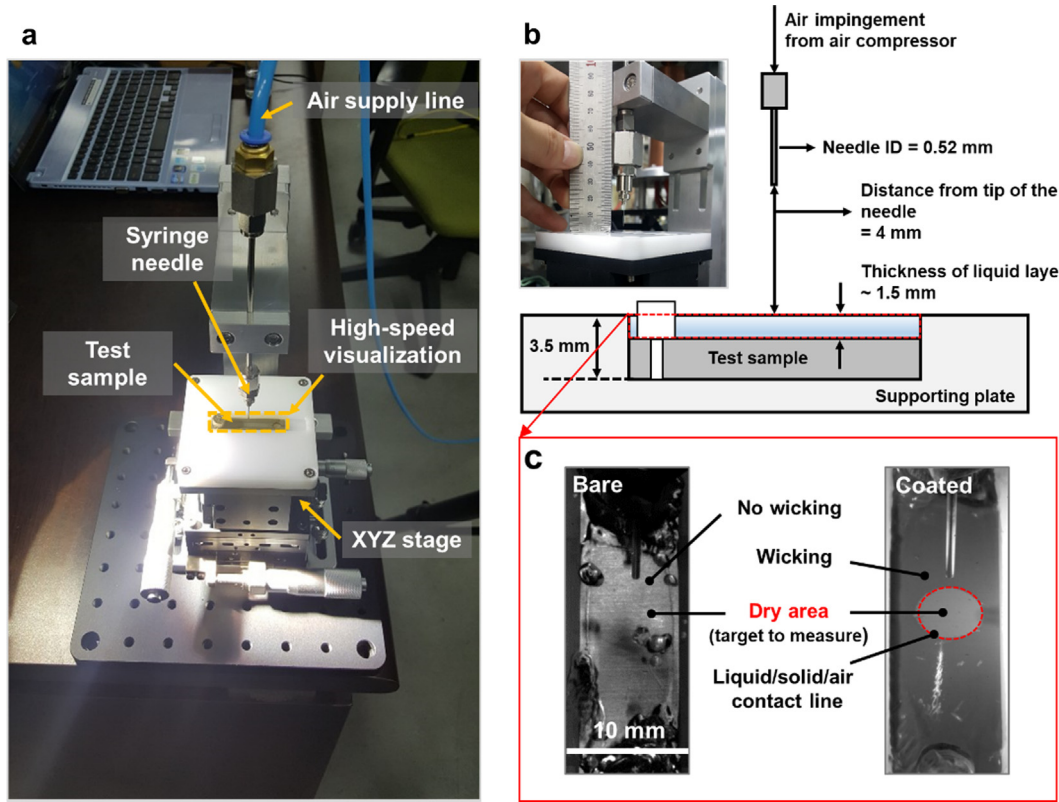


Fig. 8. (a) Photograph of the experimental apparatus and (b) schematic side-view of the test section. (c) Representative high-speed images captured 50 ms after the liquid starts to spread.

Table 2

Test matrix organized with respect to the surface characteristics, including roughness parameters, apparent contact angle, and capillary wicking capability quantified by the expansion rate of the dry area.

Sample ID	Arithmetic roughness height, R_a (μm)	Nanoscale surface area ratio, r_n (-)	Apparent contact angle, $\hat{\theta}_{sa}$ ($^\circ$)	Expansion rate of dry area, A_{dry}/dt (mm^2/ms)	Calculated dry area, A_{dry} (mm^2)
Microstructure					
B-G800	0.111 ± 0.022	1.0	63 ± 5	0.944 ± 0.115	5.00 ± 0.61
B-G600	0.146 ± 0.022	1.0	65 ± 7	0.574 ± 0.033	3.04 ± 0.17
B-G400	0.242 ± 0.014	1.0	70 ± 3	0.537 ± 0.013	2.85 ± 0.07
B-G320	0.272 ± 0.000	1.0	67 ± 6	0.543 ± 0.007	2.88 ± 0.03
Nanostructure					
Cr-M-T25	≈ 0.038	1.40 ± 0.08	≈ 0	0.855 ± 0.084	4.53 ± 0.44
Cr-M-T150	≈ 0.038	1.24 ± 0.10	≈ 0	0.944 ± 0.100	5.27 ± 0.53
Cr-M-T300	≈ 0.038	1.16 ± 0.03	≈ 0	0.653 ± 0.020	3.46 ± 0.10
Cr-M-T600	≈ 0.038	1.24 ± 0.07	≈ 0	0.796 ± 0.019	4.22 ± 0.10
Nano/microstructure					
Cr-G800-T25	0.123 ± 0.010	1.40 ± 0.08	≈ 0	0.457 ± 0.008	2.42 ± 0.04
Cr-G800-T150	0.090 ± 0.010	1.24 ± 0.10	≈ 0	0.519 ± 0.009	2.76 ± 0.05
Cr-G800-T300	0.116 ± 0.025	1.16 ± 0.03	≈ 0	0.522 ± 0.005	2.77 ± 0.02
Cr-G800-T600	0.115 ± 0.023	1.24 ± 0.07	≈ 0	0.517 ± 0.007	2.74 ± 0.04
Cr-G800-T150	0.090 ± 0.010	1.24 ± 0.02	≈ 0	0.519 ± 0.009	2.76 ± 0.05
Cr-G600-T150	0.183 ± 0.033	1.24 ± 0.02	≈ 0	0.437 ± 0.013	2.32 ± 0.07
Cr-G400-T150	0.213 ± 0.034	1.24 ± 0.02	≈ 0	0.393 ± 0.013	2.08 ± 0.06
Cr-G320-T150	0.258 ± 0.043	1.24 ± 0.02	≈ 0	0.423 ± 0.009	2.24 ± 0.05
CrAl-G800-T250	0.539 ± 0.122	1.17 ± 0.07	≈ 0	0.208 ± 0.045	1.11 ± 0.24

the test sample.

To supply a stable heat flux to the test sample made of stainless steel grade 304, Joule's direct heating method was implemented with a DC power supply (1200 A / 30 V). Based on the design of the test section, the resistance of the test sample was calculated to be $1.55 \times 10^{-3} \Omega$, which was approximately 10^3 times higher than that of the copper connector ($8.18 \times 10^{-7} \Omega$). This implies that a voltage drop occurred predominantly along the test sample rather than the copper connector, and thus, all the DC power supplied was used to generate the resistance

heat at the test sample. In this method, controlling the current input A directly regulates the voltage drop ΔV ; the heat flux was calculated by Eq. (1).

$$q'' = \frac{\text{Power}}{A_{\text{heated}}} = \frac{\Delta VI}{WL} \quad (1)$$

where W and L are the width and length of the heat transfer area, respectively. The voltage drop was measured at each end of the heated length to exclude any unintended voltage drop through the copper

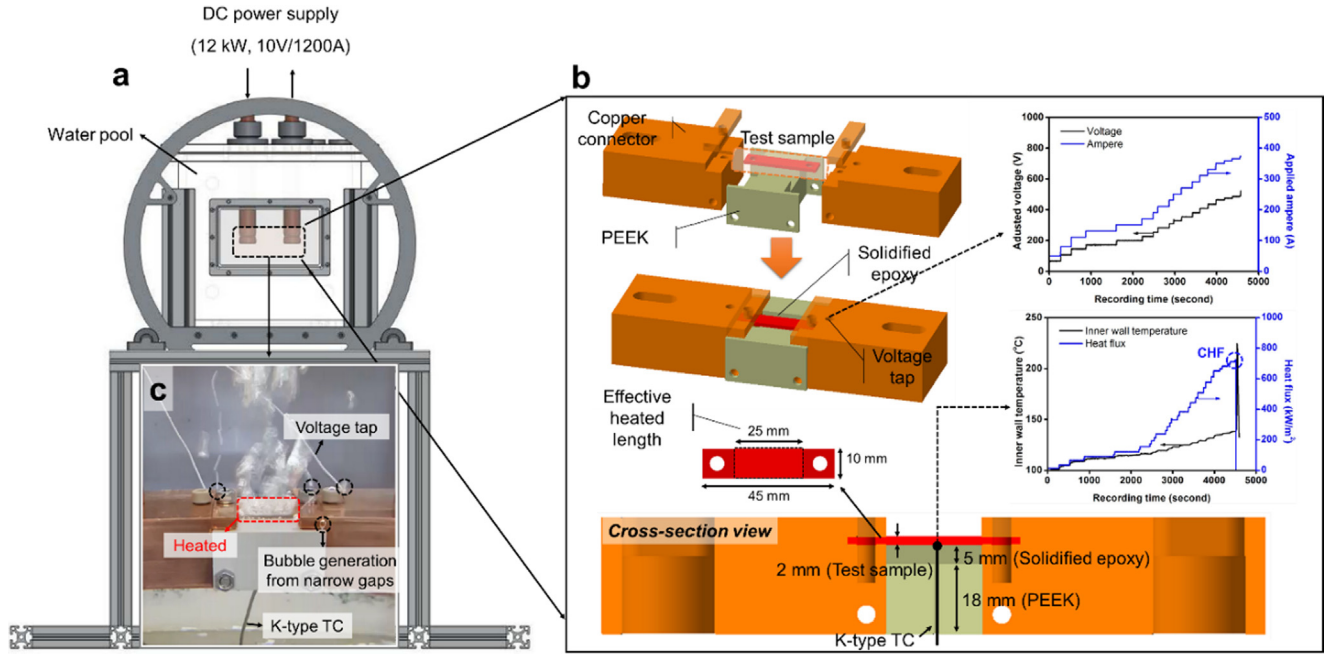


Fig. 9. (a) Pool boiling apparatus. (b) Design of the test section and the temporal readings of the voltage, ampere, inner wall temperature, and heat flux calculated using Eq. (1). (c) Visual observation of bubble formation along the test section at a heat flux of $\approx 500 \text{ kW/m}^2$, wherein no bubbles are observed over the surface of the test section except for the narrow gaps between the component parts.

electrodes (refer to the location of the voltage tap in Fig. 9(b) and (c)). As the heat flux was calculated using the four values (ΔV , I , W , and L) defined in Eq. (1), the uncertainty in the heat flux was calculated using the error propagation, as defined in Eq. (2).

$$\frac{U_{q''}}{q''} = \sqrt{\left(\frac{U_V}{V}\right)^2 + \left(\frac{U_I}{I}\right)^2 + \left(\frac{U_W}{W}\right)^2 + \left(\frac{U_L}{L}\right)^2} \quad (2)$$

The uncertainties in the voltage U_V/V , current U_I/I , width U_W/W , and heated length U_L/L were less than 0.5%, 0.5%, 2.3%, and 5.4%, respectively. Thus, the maximum measurement uncertainty of the heat flux, $U_{q''}/q''$, was calculated to be 5.9%.

The graph in Fig. 9(b) shows that the heat flux is controlled stage by stage with time, and the increments in the heat flux become smaller as it approaches the CHF. This process prevents the premature formation of the vapor film. At the occurrence of the CHF, a K-type thermocouple attached to the bottom of the test sample detected a sudden increase in the inner wall temperature over 200°C . Subsequently, to prevent the test section from a thermal attack of $> 1000^\circ\text{C}$, the DC power supply was shut down manually.

The heat loss from the test section was qualitatively evaluated under a heat flux of $\approx 500 \text{ kW/m}^2$. As shown in Fig. 9(c), no considerable bubbles are formed on the surface of the test section (e.g., copper- and PEEK-water interfaces) except for the narrow gaps between the component parts. This means that the temperature of the test section-water interface was maintained close to the saturation temperature of 100°C during the experiment. In our recent study using the same pool boiling apparatus, it was confirmed that the surface temperature of the unheated parts (i.e., PEEK and copper block) was maintained at approximately $100\text{--}103^\circ\text{C}$ with the increase in the applied heat flux. According to the 1-D heat conduction calculation, the heat loss to the unheated parts was estimated to be less than 1% of the applied heat flux. This implies that the current design for thermal insulation prevented the test section from overheating that could result in boiling-based heat loss in the water pool. Hence, although it is very difficult to perfectly prevent a potential heat loss, the current design for thermal insulation at least guarantees that the heat loss is not as much of an issue as the generation of bubbles along the unheated regions such as the copper electrodes and

PEEK.

4. Results and discussion

4.1. Nucleate boiling

In Fig. 10(a), the boiling curve shows a different trend for each surface group. The nanostructure group (Cr-M-T25/150/300/600) having a negligible R_a of $0.038 \mu\text{m}$ shows similar trends for the CHF and nucleate boiling despite a variation in r_n (1.16–1.40). No significant enhancement in boiling on the nanostructure surface is observed. In the micro/nanostructure group (Cr-G800-T25/150/300/600), by changing the grinding condition from M to G800, the CHF slightly increases with R_a (0.09–0.123 μm) although the nucleate boiling hardly changes. Upon controlling R_a up to $0.258 \mu\text{m}$ with a fixed r_n of 1.24 for the surface group Cr-G800/600/400/320-T150, it is observed that both the CHF and the nucleate boiling improves.

As shown in Fig. 10(b), no significant variation in the heat transfer coefficient (HTC) is observed with r_n (Cr-M-T25/150/300/600 and Cr-G800-T25/150/300/600); however, the HTC tends to increase with R_a (Cr-G800/600/400/320-T150). In the light of ATF surface engineering, this indicates that although controlling the substrate temperature in the sputtering process can change the nanoscale feature, its impact on the HTC is negligible. This result is consistent with our previous study [14] in which active nucleation was suppressed on Cr-coated tube surfaces having a nanostructure with $R_a \approx 0.06 \mu\text{m}$, thereby slightly lowering the HTC. Correspondingly, Li et al. [53] reported that the effect of nickel-based nanostructures (R_a : $0.021\text{--}0.085 \mu\text{m}$, r_n : 1.11–1.45) on the HTC was negligible and might be worse compared to a bare surface owing to reductions in the nucleate site density as well as the bubble departure diameter and bubble departure frequency. Witharana et al. [54] experimentally validated that the wall superheat required for active nucleation increases as the diameter of the artificial cavity decreases from micro- to nanoscale (5000–50 nm); this variation agreed with the Young-Laplace equation: $P_v - P_l = \frac{2\sigma_{lv}}{r_c}$. These results support the viewpoint that controlling r_n without the microscale features is ineffective on the HTC because it rarely activates nucleation on the nanosized cavities.

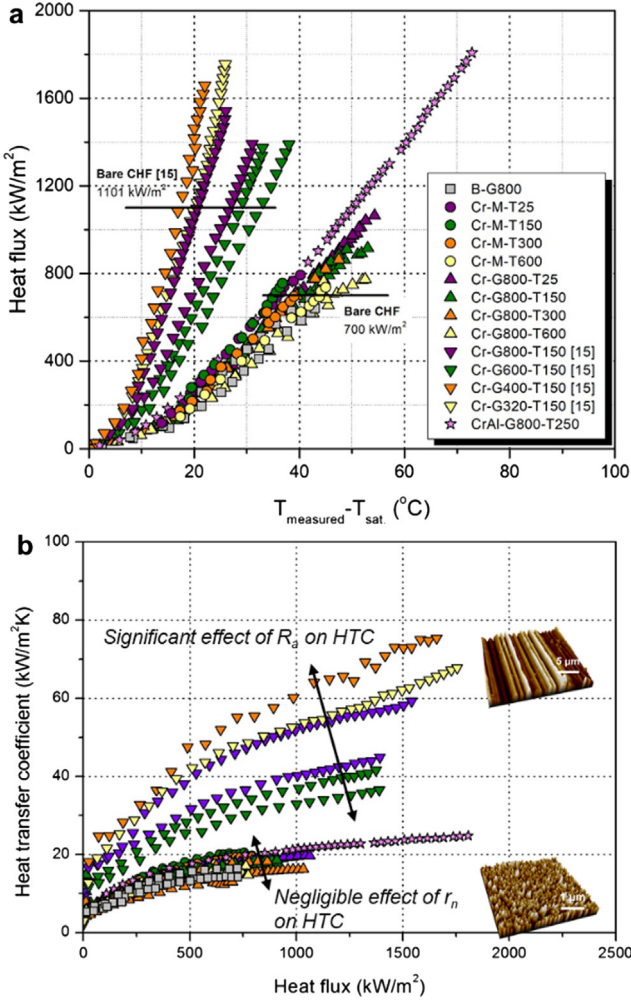


Fig. 10. (a) Boiling curve and (b) heat transfer coefficient. The boiling curves of B-G600/400/320 are not presented here; however, the CHF values are presented in Fig. 12. Moreover, it should be noted that a direct comparison of the HTC between [15] and the present study is limited due to a difference in the bare CHF. Thus, only a relative comparison of the HTC trends for each study is possible. All the boiling curves are also presented in our relevant study [51].

Contrary to the ineffectiveness of r_n on the HTC, the HTC tends to change as R_a exceeds approximately 0.1 μm (Fig. 11(a)). Similarly, Kim et al. [24] observed that the HTC increased with R_a , the range of which (0.04–2.35 μm) was within the current roughness range (0.11–0.54 μm). This result is also consistent with that obtained by Dadjoo et al. [55]. They observed that the nucleate boiling efficiency was higher on a ground surface than on a polished surface; moreover, the difference between the HTC became pronounced with the pool boiling of the silica nanofluid. This indicates that grinding the surface before either fabricating nanostructures or depositing nanoparticles is more effective on the HTC.

To predict the effect of the surface parameters on the wall heat flux q_w , Li et al. [56] modified Rohsenow's semi-empirical correlation [57] with a surface influencing factor C_s . The modified correlation is defined as follows:

$$\frac{c_{pl}\Delta T_{\text{sat}}}{h_{fg}} = 0.013C_s^{-0.33} \left[\frac{q_w}{h_{fg}\mu_l} \sqrt{\frac{\sigma}{g(\rho_l - \rho_v)}} \right]^{0.33} Pr_l \quad (3)$$

$$C_s \gamma^{0.04} = (1 - \cos\theta)^{0.5} \left[1 + \frac{5.45}{(Ra - 3.5)^2 + 2.61} \right] \quad (4)$$

where $\theta = \text{MAX}(\theta, 15^\circ)$ and γ is the influence parameter of the heating

surface material. For a particular heating material, Eq. (4) varies according to θ and R_a . Considering that all the values of θ_a on the present coated surfaces are close to 0° , Eq. (4) predicts the nonmonotonic trend of $C_s \gamma^{0.04}$ with R_a , as shown in Fig. 11(b). In particular, the trend shows that the HTC increases with R_a before attaining the maximum heat transfer rate at $R_a = 3.5 \mu\text{m}$, after which it starts to decline. As the measured roughness values ($\approx 0.6 \mu\text{m}$) are significantly lower than the maximum value (3.5 μm), according to Eq. (4), it is reasonable that the HTC increases with R_a within the current roughness range. Nonetheless, it should be noted that since Eq. (4) neglects the effect of r_n on the surface influencing factor, i.e., nucleate boiling, the actual variation of the HTC on surfaces having a nanoscale feature is not fully consistent with R_a . This limits a detailed description of the HTC trend. Therefore, although a more detailed analysis on the HTC with surface characteristics is beyond the scope of this study, the coupled effect of r_n and R_a on the HTC needs to be further investigated in the future.

4.2. Critical heat flux

Fig. 12 compares the CHF enhancement ratio of the surfaces coated with the ATF candidate materials, including Cr [10,13–15], FeCrAl [12,14], ZrSi₂ [11], and the present data. In Fig. 12(a), it is observed that the CHF is enhanced as the surface roughness increases, regardless of the surface coating. Most of the CHF data reported till date show an enhancement up to approximately 50% within the roughness range of 0.3 μm. Although more enhancement in the CHF can be achieved if one further increases the surface roughness above 0.3 μm, it might be unfavorable for the oxidation resistance of the coated surface. Note that the roughness value of 0.3 μm used in this study is not a technical standard but a typical engineering value obtained from conventional nuclear fuel claddings; thus, the value can be changed according to different engineering or regulatory points of view.

Focusing on the trend of the present data, the CHF enhancement trend in Fig. 12(b) can be separated into three groups based on the surface engineering processes, i.e., grinding, sputtering, grinding and sputtering. The extent of the enhancement is lower in the sputtered surfaces than in the ground and sputtered surfaces. Grinding the surface before sputtering effectively enhances the CHF beyond the upper bound of the CHF enhancement achievable on the sputtered surfaces. In contrast, when the surface is sputtered by Cr without a grinding process, the CHF is marginally enhanced up to 9%, regardless of the variation in r_n . Considering the extreme thermal-hydraulic conditions in the nuclear core, a CHF enhancement of 9% is very low. Therefore, it would be

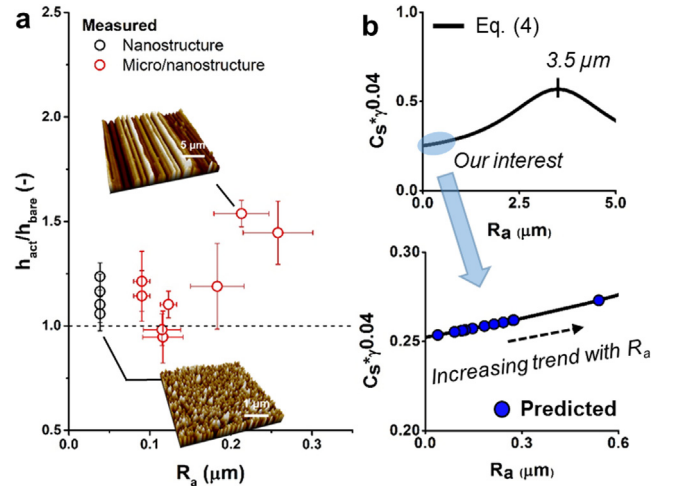


Fig. 11. (a) Variation of the HTC with R_a at a heat flux of 500 kW/m². In the y-axis, $h_{\text{act}}/h_{\text{bare}}$ denotes the ratio of the actual HTC to the bare HTC. (b) Effect of R_a on the surface-influenced nucleate boiling efficiency, $C_s \gamma^{0.04}$, on a super-hydrophilic surface ($\theta_a = 0^\circ$).

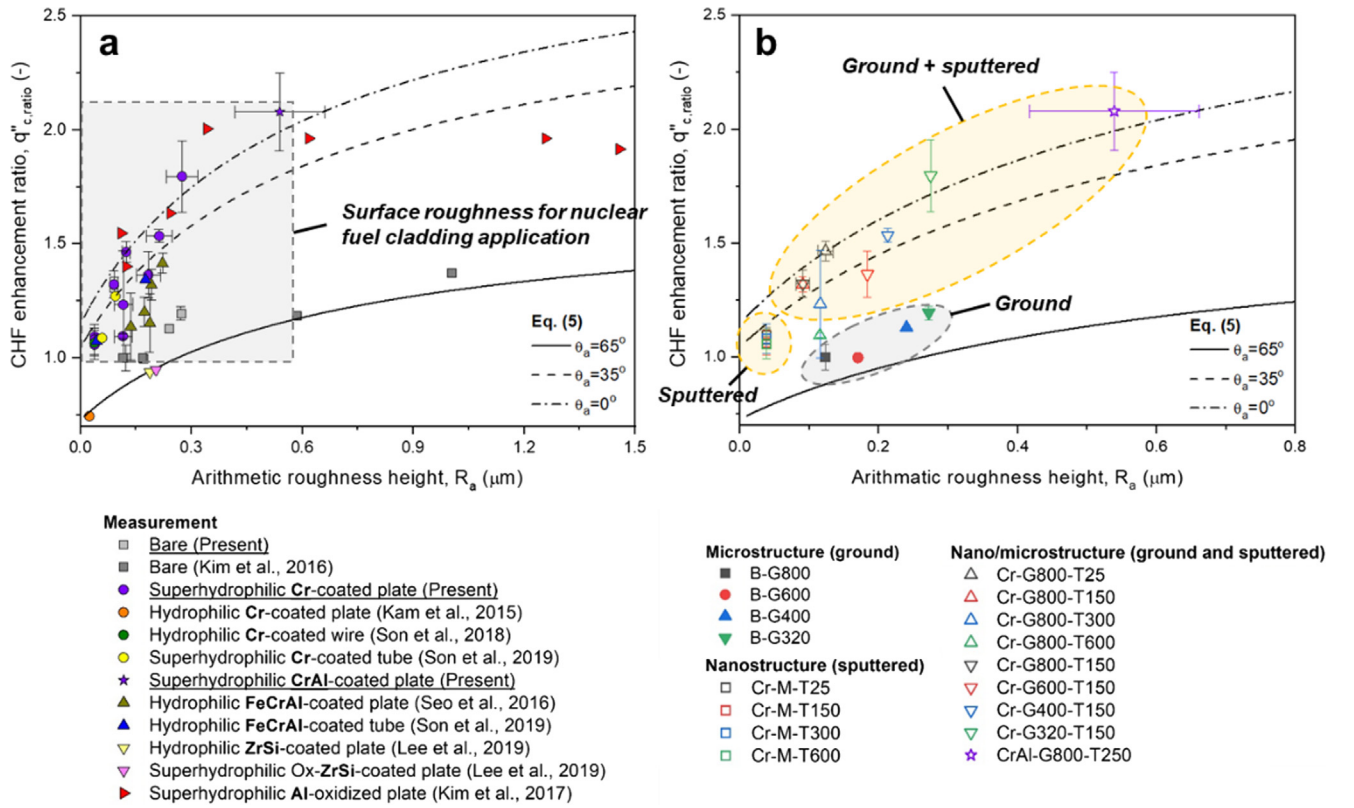


Fig. 12. Increasing trend of the CHF enhancement ratio with surface roughness: (a) ATF-functional engineered surfaces reported till date and (b) present surfaces.

difficult to achieve a meaningful DNB gain in the ATF-functional claddings using the sputtering process exclusively without the grinding process.

The trend of the CHF has been predicted by Kim et al. using a roughness- and wettability-based CHF model [24]. They considered that the capillary wicking force along the grooved channels additionally resisted the vapor recoil force. Assuming that the grooved channel conceptually corresponds to the capillary tube, they modified Kandlikar's wettability model [20] such that the capillary wicking force in the last term in the right-hand side was proportional to the surface roughness ratio R_a/S_m . In this case, S_m denotes the mean spacing between the roughness peaks.

$$q''_c = S \times h_{fg} \rho_v^{\frac{1}{2}} [\sigma_{lv} g (\rho_l - \rho_v)]^{\frac{1}{4}} \times \frac{(1 + \cos\theta)}{16} \left[\frac{2}{\pi} + \frac{\pi}{4} (1 + \cos\theta) + \frac{4C \cos\theta}{1 + \cos\theta} \left(\frac{R_a}{S_m} \right) \right]^{1/2} \quad (5)$$

By fitting the measured data including the contact angle, roughness, and CHF into Eq. (5), they determined the unknown constants of S and C as 0.811 and 87.9, respectively. Moreover, the roughness parameters, R_a and S_m , are practically correlated to each other because a sandpaper consists of spherical particles having an identical size, which ensures a particular relation between the peak-to-peak and peak-to-valley distances along the roughness profile. Using the roughness data reported in [24], we infer that the relation follows S_m (μm) = 13.3 × R_a .

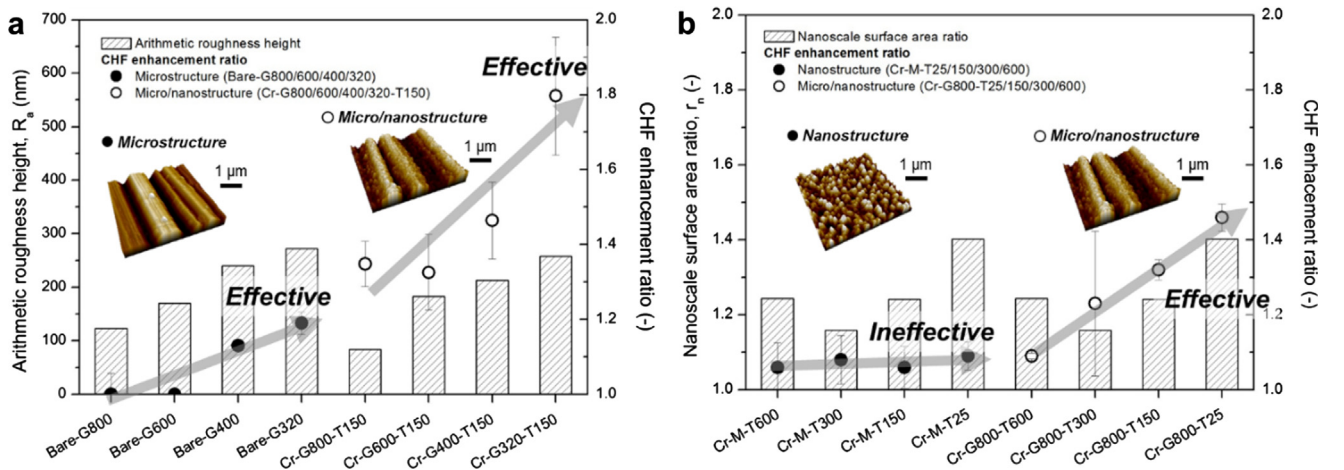


Fig. 13. Effects of (a) arithmetic roughness height R_a and (b) nanoscale surface area ratio r_n on the CHF.

(μm) + 14; this equation enables us to predict the CHF with a single roughness value of R_a . As shown in Fig. 12(a), Eq. (5) generally predicts the trend of the CHF enhancement ratio with surface roughness even in the current roughness range (Fig. 12(b)). Based on this trend, it is discerned that the surface roughness R_a is of paramount importance to enhance the CHF of ATF-functional surfaces. Besides the strong effect of R_a on the CHF, it is observed that some of the CHF data are scattered from the prediction line, which can be attributed to the variation in the nanoscale surface area ratio r_n . It should be noted that, at the same R_a and r_n , it is difficult to expect that the CHF enhancement is due to slight differences in the intrinsic contact angle ($50\text{--}80^\circ$), thermal property, and microscale thickness of the coated layer ($10^0\text{--}10^2\ \mu\text{m}$) between the metallic ATF materials.

To understand the role of micro- and nanostructures on the CHF, three types of the CHF enhancement trend are compared with the surface parameters of R_a , r_n , and its combination in Fig. 13. The effect of R_a on the CHF is shown in Fig. 13(a). On the ground surfaces (Bare-G800/600/400/320), a decrease in the grit number of sandpaper, which increases R_a at microscale, results in an enhancement of the CHF by 19%. This minimal enhancement has been typically observed in case of hydrophilic ground or polished metal surfaces. In numerous experimental observations [15,23,24], it was observed that a higher surface roughness augments the length of the triple contact lines [23] and even activates capillary wicking [15,24], thereby enhancing the CHF monotonically with surface roughness. Thus, with a lower grit number of the sandpaper, a higher R_a and a higher CHF enhancement are achieved.

The effect of r_n on the CHF is shown in Fig. 13(b). When the coating materials are deposited on a mirror-polished surface having $R_a = 0.038\ \mu\text{m}$ with varied substrate temperatures (Cr-M-T25/150/300/600), the CHF demonstrates a minimal enhancement of up to 9%. Moreover, a small variation of 6–9% in the CHF is observed as r_n increases from 1.16 to 1.42. Similarly, Son et al. [13] observed a minimal enhancement of 6% on a Cr-coated surface despite increases in r_n and

wettability. These results imply that the effect of r_n on the CHF is almost negligible if microscale features are not present in the surface. Indeed, r_n starts to affect the CHF, when an intrinsic microstructure is provided by surface grinding (Cr-G800-T25/150/300/600). In this case, the CHF is enhanced from 9 to 46% for a similar variation in r_n (it increases from 1.16 to 1.42). Therefore, it is evident that although a lower substrate temperature in the sputtering process generates a higher r_n , its effect on the CHF depends on the existence of microscale roughness.

Lastly, the combination of R_a and r_n creates a synergistic effect on the CHF, which is shown in Fig. 13(a). When a fixed r_n of 1.24 is applied to differently ground surfaces (Cr-G800/600/400/320-T150), a variation in R_a from 0.090 to 0.258 μm enhances the CHF by up to 79%. Moreover, in case of CrAl-G800-T250, the highest R_a value of 0.539 μm leads to further enhancement of the CHF (110%). Considering the relationship between the surface structure and CHF, it is emphasized that the surface coating process must be conducted on a ground surface and not on a mirror surface in the light of improving the thermal safety of the ATF cladding.

4.3. Analysis of CHF enhancement with capillary wicking phenomenon

According to the previous findings that encompass surface characterization [32,47,49], *in-situ* observation of a liquid/vapor distribution [58], and corresponding theoretical models [23,24,59,60], it has been established that one of the key mechanisms enhancing the CHF on roughness-controlled surfaces originates from the capillary wicking phenomenon, which is capable of passively supplying the cooling liquid responsible for additional evaporation. In particular, the mechanism implies that the longer the imbibition length of the capillary flow, the higher is the liquid area fraction over the boiling surface so that the CHF can be enhanced with the corresponding portion of the liquid area fraction. Based on this demonstration, we analyze an underlying cause of enhancing the CHF on roughness-controlled surfaces through the experimental observation of the growth behavior of the dry spot

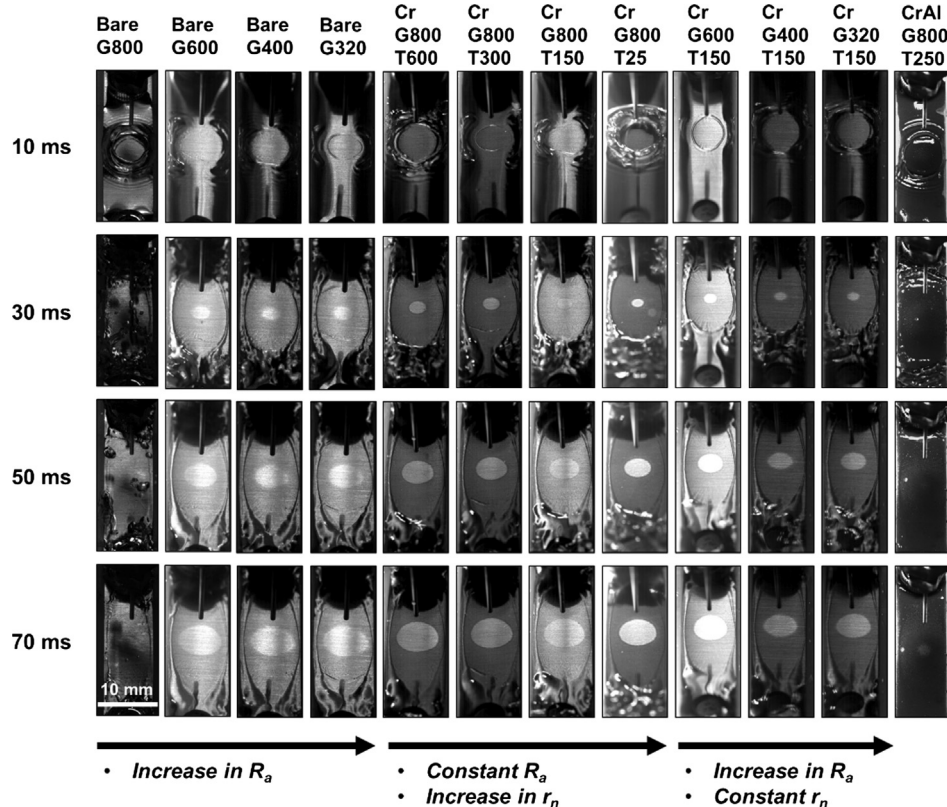


Fig. 14. Visual observation of a growing dry area on the fabricated surfaces.

characterized in Section 3.2.

Fig. 14 shows that although a dramatic variation in A_{dry} with R_a is not observed for a narrow range of R_a (0.1–0.5 μm), A_{dry} tends to gradually decrease as R_a increases. Moreover, an increase in r_n in addition to R_a contributes to a smaller A_{dry} because the nanostructure augments the liquid/solid contact line, thereby assisting the liquid to adhere to the surface [33]. In particular, a significant delay in the liquid receding is observed for CrAl-G800-T250, owing to its two times higher R_a compared to the others. Based on this observation, it is noteworthy that augmenting the surface roughness and adding the nanoscale feature on the rougher surface are beneficial for further enhancing the capillarity, which, in turn, provides an increased resistance to recede from the expansion of the dry area. Furthermore, in comparison with the CHF (Fig. 15), it is observed that the decreasing trend of A_{dry} with R_a is inversely proportional to an increase in the CHF. This correlation emphasizes that the size of the dry spot, controlled by the surface capillarity, is significantly related to triggering the CHF. Under a pool boiling condition, Park et al. [61] likewise observed that a reduction in the bubble base diameter on structured surfaces is favorable for the successful rewetting and corresponding enhancement of CHF. Considering the similarity of the result, we infer that the capillary wicking behavior characterized in this study has a potential linkage to the actual behavior of bubble dry spots.

To link A_{dry} estimated under a room adiabatic condition to a liquid/vapor phase distribution under actual boiling conditions, we first simplify a near-CHF boiling structure on a rough surface as shown in Fig. 16(a). According to the *in-situ* observations of the boiling structure using infrared (IR) thermography [62] and synchrotron X-ray imaging [58], the liquid/vapor phase distribution underneath the boiling bubbles merging with each other can be depicted by separating a bubble dry spot/patch and the surrounding wicked liquid inside a given bubble base meniscus (Fig. 16(b)). Then, the liquid area fraction, $1-\alpha$, with the bubble base area, A_b , is simply expressed as $(A_b - A_{ds})/A_b$, where A_{ds} denotes the area of the bubble dry spot. Herein, A_{ds} is replaced by A_{dry} , assuming that the hydrodynamic behavior of the bubble dry spot (A_{ds}) on the rough surface is analogous to the present observation of the area-based capillary wicking (A_{dry}). Moreover, assuming that the bubble base diameter corresponds to the capillary length $L_c = \sqrt{\sigma_{lv}/\rho_l g} = 2.5\text{mm}$ for saturated water, A_b is calculated to be 4.85 mm^2 . This assumption is acceptable according to Kim et al. [63]'s observation, where the size of the bubble departing from the structured surfaces ranges from 2 to 3 mm when approaching the CHF, even though it demonstrates significant variations with respect to the surface characteristics at low and moderate heat flux regions where isolated bubbles occur predominantly [61]. Kim et al. [62] further explained that on structured surfaces, the movement of the triple contact line remains unchanged as the bubble grows due to the pinning effect caused by an increase in the liquid/vapor frictional force along the triple contact line.

The $1-\alpha$ calculated with the provided values of A_b and A_{dry} shows a linear relation with the CHF enhancement ratio (Fig. 16(c)). This relation is supported by Jung et al. [52]'s observation, in which the CHF is triggered at the moment when the increasing rate of the dry patch is beyond the evaporation rate through the liquid-occupied region. It implies that delaying the growth of the dry area enables the surrounding liquid absorbed more to transfer the heat as much as CHF gain. Moreover, our result is consistent with those obtained on structured surfaces [61,64], in which the *in-situ* liquid-occupied area and liquid area fraction were measured directly using an infrared-based detection technique. In particular, assuming that the heat transfer mechanism along the wicked liquid is governed by thermal conduction inside a liquid film having a few micrometers thickness, Kim et al. [64] described the relation between the CHF and the liquid area fraction as

$$q_c'' \approx \frac{k_l \Delta T_c}{\delta_l} (1 - \alpha). \quad (6)$$

At T_{crit} , the dry spot is not rewetted again, leading to the triggering of the CHF. For a given T_{crit} , Eq. (6) predicts a linear correlation between the CHF and the liquid area fraction; this trend is also observed in Fig. 16(c). By numerically solving a heat diffusion equation in the dry spot region, Kwak et al. [65] inferred that a larger size of the dry spots generates excessive heat inside the dry spots, which results in reaching the critical temperature faster. Based on the fact that the CHF is strongly correlated to both the size of the dry spot and the critical temperature, its thermohydrodynamic relation is estimated as

$$q_c'' \propto \frac{T_c}{D_{dry}^2} \propto \frac{T_c}{A_{dry}} \quad (7)$$

which predicts an inverse relation between A_{dry} and the CHF. Thus, considering the good agreement between the present data and the predictions of Eqs. (6) and (7), it can be qualitatively inferred that a higher capillarity on a rougher surface causes more imbibition of the rewetting liquid along the triple contact lines, and this process tends to resist the growth of the dry spot. Consequently, the reduction in the size of the dry spot and the increase in the liquid area fraction induce the formation of reversible dry spots that are not overheated so much as to trigger the CHF.

4.4. Discussion and recommendations for future study

Based on the current observation and useful insights from recent studies, it is inferred that the CHF is significantly affected by the growth behavior of the dry spot resulting from the capillary wicking capability of engineered surfaces. Accordingly, the key to enhance the CHF of an engineered ATF-functional surface is to determine ways to delay the growth of the dry spots by controlling the surface roughness at nano/microscale. In our experiment, no CHF maxima are observed within $R_a \approx 0.5\text{ }\mu\text{m}$; the dry area consistently decreases as the surface roughness increases at least within the roughness range for ATF cladding applications. This provides mounting evidence that a higher surface roughness would be beneficial for achieving higher CHF in an ATF-functional surface, even though we have only observed it under a pool boiling condition. Moreover, considering that the dry area is smaller on a hierarchical structure (micro/nanostructure) than on independent structures (microstructure and nanostructure), this result further emphasizes the practical importance of hierarchically engineering the surface at nano/microscale to improve the thermal safety of the ATF cladding.

Although this study suggests a design basis for clarifying the role of the surface feature on the CHF of ATF claddings, the practical usage of the result would be limited to a laboratory-scale condition of atmospheric-pressure pool boiling on fresh-engineered surfaces. This is because the extreme thermal-hydraulic and water chemistry

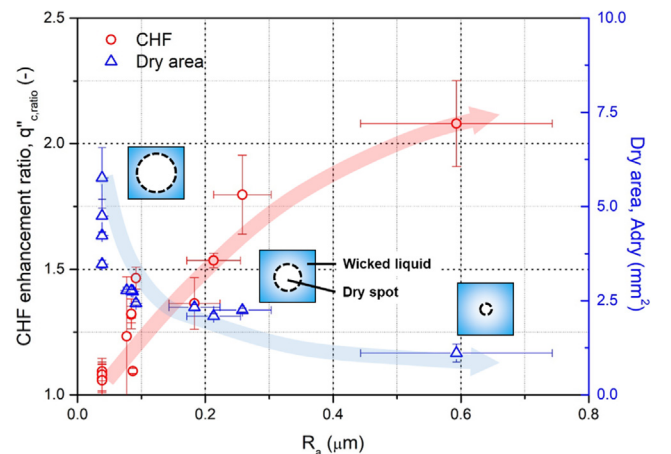


Fig. 15. Inversely proportional relation between A_{dry} and CHF.

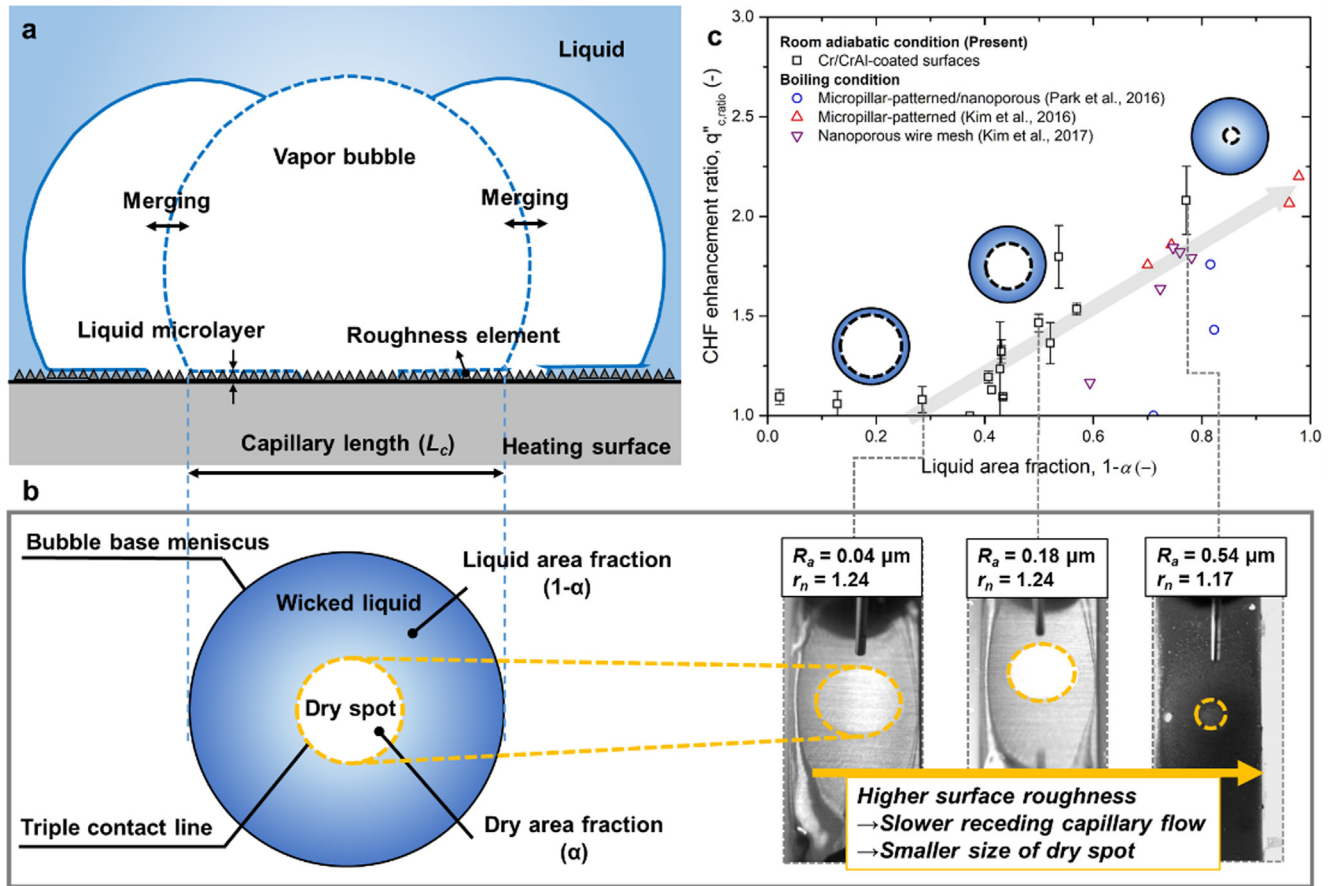


Fig. 16. (a) Schematic of the boiling bubbles on a rough surface. (b) Bottom view of a dry spot/wicked liquid structure and corresponding observation of the receding capillary flow varying according to the surface roughness. (c) Proportional relation between the liquid area fraction and CHF enhancement ratio.

environments in nuclear reactor cores mostly require high-pressure flow boiling on aged-engineered surfaces. Moreover, since an RIA leading to a sudden excursion in the thermal power alters the boiling curve and even the CHF [66], it is necessary to further consider a *transient* CHF rather than a *steady-state* CHF on the ATF cladding. The flow boiling CHF experiment coupled with the ATF candidate materials is notably lacking. Very recently, Lee et al. [25] reported that the steady-state and transient flow boiling CHF of an FeCrAl alloy is affected not by the surface characteristics but by the thermal property. However, since they used the FeCrAl alloy as a body material rather than systematically controlling its surface morphology at nano/micro-scale, the effect of the surface characteristics of the ATF claddings on the flow boiling CHF is not understood yet.

In terms of surface aging, a few studies have shown that the oxidation of the ATF-functional surfaces either preserves or even enhances

the CHF [9,11,67] and the quench performance [17,19]. Moreover, it has been confirmed that a nanoscale morphology of the Cr-coated surface is resistant to clogging during oxidation under 400 °C dry air [67]; thus, nanotechnology can be applied to the surface engineering of the ATF claddings. Nonetheless, under the water chemistry conditions in the primary loop of a nuclear power plant, the surface characteristics of the ATF cladding are inevitably affected by the cumulative deposition of Chalk River unidentified deposits (CRUD) in the region where the subcooled boiling occurs [6]. Considering all the aforementioned limitations, future studies need to evaluate the realistic effects of core thermal-hydraulic conditions (*i.e.*, high-pressure flow boiling), accident conditions (*i.e.*, RIA), and water chemistry (*i.e.*, oxidation and CRUD formation) on the thermal safety of the ATF cladding.

Table 3

Effects of the fabrication methods and the resultant surface characteristics on the pool boiling performance.

Fabrication methods	Types of surface structure	Surface characteristics	Pool boiling performance	
			Nucleate boiling	CHF enhancement
DC sputtering	Nanostructure (Fig. 2(a))	$R_a = 0.04 \mu m$ $r_n = 1.16-1.40$ $\theta_a \approx 0^\circ$	Ineffective	$\approx 9\%$
Grinding	Microstructure (Fig. 2(b))	$R_a = 0.11-0.27 \mu m$ $r_n = 1.0$ $\theta_a = 60-70^\circ$	n/a	$\approx 19\%$
Grinding and DC sputtering	Nano/microstructure (Fig. 2(c))	$R_a = 0.90-0.26 \mu m$ $r_n = 1.16-1.40$ $\theta_a \approx 0^\circ$	Effective	$\approx 79\%$

5. Conclusions

This study investigated the CHF of an ATF-functional surface by controlling the surface roughness at nano/microscale. To diversify the roughness value and scale, we fabricated a total of 17 roughness-controlled surfaces (four ground surface, twelve Cr-coated surfaces, and one CrAl-coated surface). Based on the measurements of the basic surface characteristics (R_a , r_n , and θ_a), capillary wicking capability (A_{dry}), and boiling parameters (HTC and CHF) of the surfaces, the effect of the nano/microscale roughness of the ATF-functional surface on the CHF was analyzed. The major findings are summarized as follows.

- (1) As presented in Table 3, it was difficult to improve the nucleate boiling heat transfer by grinding or sputtering the surface exclusively. In contrast, hierarchically augmenting the surface structure resulted in a synergetic effect on the CHF as well as the nucleate boiling efficiency. The CHF showed an increasing trend within the current range of surface roughness (0.11–0.54 μm) without any maxima.
- (2) Importantly, enlarging the nanoscale surface area without the microscale feature was ineffective in enhancing the CHF. To resolve this limitation at least under the pool boiling condition, the apparent microstructure should be guaranteed an R_a of $> 0.1 \mu\text{m}$. Therefore, from the perspective of thermal engineering, depositing the ATF materials onto the smooth surface with a variation in the nanoscale features (not porous) can be ineffective in improving the thermal efficiency and safety of the nuclear fuel claddings.
- (3) Nevertheless, the nanoscale feature assisted by nuclear fuel cladding-level surface roughness ($R_a \approx 0.3 \mu\text{m}$) could generate a capillary action, which was as much as the CHF enhancement. From the underlying relation between them, it was confirmed that a higher surface roughness and a more delayed growth in the dry area could result in a higher liquid area fraction, which was responsible for the additional evaporation of the wicked liquid inside the surface structures. The correlation between the dry area, the liquid area fraction, and the CHF was consistent with those reported previously.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2019.114786>.

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